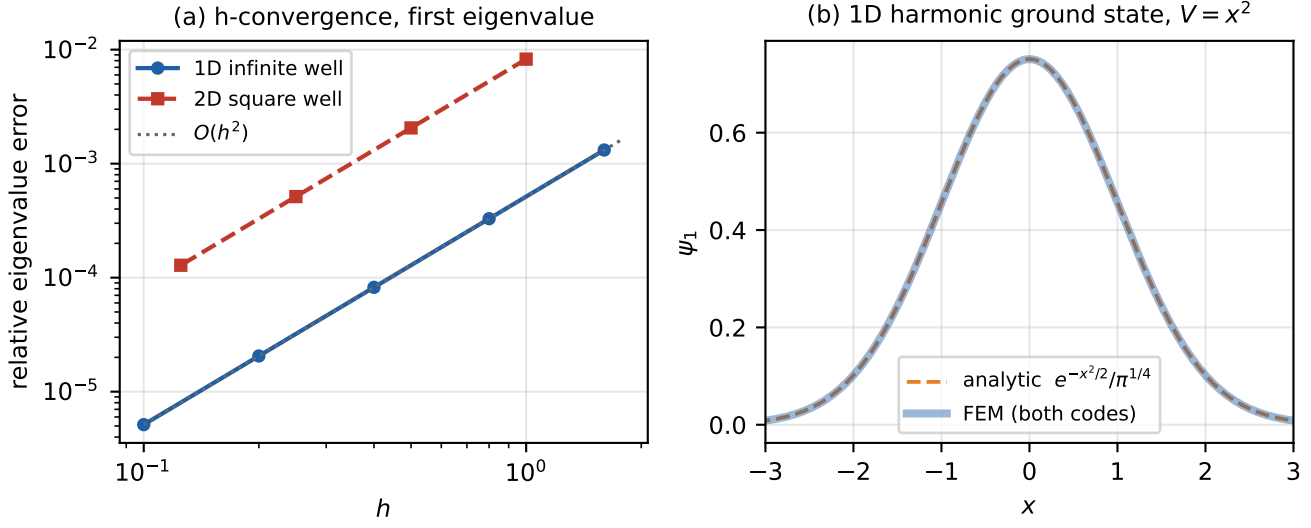
(a) h-convergence, first eigenvalue
relative eigenvalue error
10^{-2}
10^{-3}
10^{-4}
10^{-5}
1D infinite well
2D square well
$O(h^2)$
10^{-1}
10^{0}
h
(b) 1D harmonic ground state, $V = x^2$
ψ_1
0.6
0.4
0.2
0.0
analytic $e^{-x^2/2}/\pi^{1/4}$
FEM (both codes)
−3
−2
−1
0
1
2
3
x